Exercise 16.2

1. Examine whether each of the following system of equations is diagonally dominant:

a) $3x_1 + x_2 = 7, x_1 - 3x_2 = 9$

Sol^n: Here, $3x_1 + x_2 = 7$ ... (i)

$x_1 - 3x_2 = 9$ ....(ii)

$a_{11} = 3, a_{12} = 1$ such that $|a_{11}| > |a_{12}|$

$a_{21} = 1, a_{22} = -3$ such that $|a_{22}| > |a_{21}|$

$\therefore$ it is diagonally dominant and can solve the system using gauss-Seidel method

So, $x_1 = \frac{7 - x_2}{3}$ and $x_2 = \frac{x_1 - 9}{2}$

Initially, set $x_2 = 0$ to get the successive iteration followed by

Iteration I: $x_1^1 = \frac{7}{3} = 2.3333$

$x_2^1 = -2.2222$

Iteration 2: $x_1^2 = \frac{7 + x_2}{3} = \frac{7 + 2.2222}{3} = 3.0741$

$x_2^2 = \frac{x_1 - 9}{2} = \frac{3.0741 - 9}{2} = -1.9753$

Also, absolute relative error formula is given by

$$E(x_n) = \left| \frac{x_{n+1} - x_n}{x_{n+1}} \right| \times 100\%$$

$\therefore$ The absolute relative error of the solution of the system in second order is

$$E(x_1) = \left| \frac{x_1^2 - x_1^1}{x_1^2} \right| \times 100\% = \left| \frac{3.0741 - 2.3333}{3.0741} \right| \times 100\% = 24.098\% \approx 24.1\%$$

$$E(x_2) = \left| \frac{x_2^2 - x_2^1}{x_2^2} \right| \times 100\% = \left| \frac{-1.9753 + 2.2222}{-1.9753} \right| \times 100\% = 12.499\% \approx 12.5\%$$

b) $2x_1 + 5x_2 = 7, 5x_1 - 2x_2 = 2$

Sol^n: Here, $2x_1 + 5x_2 = 7$ ... (i)

$5x_1 - 2x_2 = 2$ ....(ii)

$a_{11} = 2, a_{12} = 5$ such that $|a_{11}| \not> |a_{12}|$

$a_{21} = 5, a_{22} = -2$ such that $|a_{22}| \not> |a_{21}|$

$\therefore$ it is not diagonally dominant and cannot solve the system using Gauss-Seidel method

The system can be made diagonally dominant if it is rewrite as

$5x_1 - 2x_2 = 2$ .... (iii)

$2x_1 + 5x_2 = 7$ ..... (iv)

Such that $|a_{11}| > |a_{12}|$

$|a_{22}| > |a_{21}|$

$\therefore$ System is diagonally dominant, we can apply Gauss-Seidel method. Therefore the system is rewrite as $x_1 = \frac{2+2x_2}{5}$ and $x_2 = \frac{4-2x_1}{5}$

Initially, set $x_2 = 0$ to get the successive iteration followed by

Iteration I: $x_1^1 = \frac{2}{5} = 0.4000$

$$x_2^1 = \frac{4 - 2 \times 0.4}{5} = 0.6400$$

Iteration 2: $x_1^2 = \frac{2+2 \times 0.6400}{5} = 0.6560$

$$x_2^2 = \frac{4 - 2 \times 0.6560}{5} = 0.5376$$

Also, absolute relative error formula is given by

$$E(x_n) = \left| \frac{x_{n+1} - x_n}{x_{n+1}} \right| \times 100\%$$

$\therefore$ The absolute relative error of the solution of the system in second order is

$$E(x_1) = \left| \frac{x_1^2 - x_1^1}{x_1^2} \right| \times 100\% = \left| \frac{0.6560 - 0.4000}{0.6560} \right| \times 100\% = 39.024\%$$

$$E(x_2) = \left| \frac{x_2^2 - x_2^1}{x_2^2} \right| \times 100\% = \left| \frac{0.5376 - 0.6400}{0.5376} \right| \times 100\% = 19.047\%$$

c) $3x + y + z = -1$, $2x + 5y + z = 5$, $x + 2y - 4z = 8$

Sol^n: Here, $a_{11} = 2$, $a_{12} = 5$ and $a_{13} = 1$ such that $|a_{11}| > |a_{12}| + |a_{13}|$

$a_{21} = 2$, $a_{22} = 5$ and $a_{23} = 1$ such that $|a_{22}| > |a_{21}| + |a_{23}|$

$a_{31} = 1$, $a_{32} = 2$ and $a_{33} = -4$ such that $|a_{33}| > |a_{31}| + |a_{32}|$

$\therefore$ it is diagonally dominant can solve the system using Gauss-Seidel method. Therefore, the system is rewrite as $x = \frac{-1-y-z}{3}$, $y = \frac{5-2x-z}{5}$ and $z = \frac{-8+x+2y}{4}$

Initially, set $y = z = 0$ to get the successive iteration followed by

Iteration I: $x = \frac{-1-0-0}{3} = -0.3333$

$$y = \frac{5 - 2 \times (-0.3333) - 0}{5} = 1.1333$$

$$z = \frac{-8 - 0.3333 + 2 \times 1.1333}{4} = -1.5167$$

Iteration 2: $x = \frac{-1 - 1.1333 + 1.5167}{3} = -0.2056$

$$y = \frac{5 - 2 \times (-0.2056) + 1.5167}{5} = 1.3856$$

$$z = \frac{-8 - 0.2056 + 2 \times 1.3856}{4} = -1.3586$$

Also, absolute relative error formula is given by

$$E = \left| \frac{x_{n+1} - x_n}{x_{n+1}} \right| \times 100\%$$

$\therefore$ The absolute relative error of the solution of the system in second order is

$$E(x) = \left| \frac{x_2 - x_1}{x_2} \right| \times 100\% = 0.6211\%$$

$$E(y) = \left| \frac{y_2 - y_1}{y_2} \right| \times 100\% = 18.2\%$$

$$E(z) = \left| \frac{z_2 - z_1}{z_2} \right| \times 100\% = 11.636\%$$

d) $-x + y + 3z = 3, 3x - 6y - 2z = 4, 5x + 3y + z = 4$

Sol^n: Here, $a_{11} = -1$, $a_{12} = 1$ and $a_{13} = 3$ such that $|a_{11}| \not> |a_{12}| + |a_{13}|$

$a_{21} = 3$, $a_{22} = -6$ and $a_{23} = -2$such that $|a_{22}| > |a_{21}| + |a_{23}|$

$a_{31} = 5$, $a_{32} = 3$ and $a_{33} = 1$such that $|a_{33}| \not> |a_{31}| + |a_{32}|$

$\therefore$ it is not diagonally dominant cannot solve the system using Gauss-Seidel method. The system can be made diagonally dominant if it is rearranged as

$5x + 3y + z = 4, 3x - 6y - 2z = 4, -x + y + 3z = 3$

Such that $|a_{11}| = 5 > |a_{12}| + |a_{13}| = 3 + 1 = 4$

$|a_{22}| = 6 > |a_{21}| + |a_{23}| = 3 + 2 = 5$

$|a_{33}| = 3 > |a_{31}| + |a_{32}| = 1 + 1 = 2$

$\therefore$ System is diagonally dominant, we can apply Gauss-Seidel method. Therefore the system is rewrite as $x = \frac{4 - 3y - z}{5}$, $y = \frac{-4+3x - 2z}{6}$ and $z = \frac{3 + x - y}{3}$

Initially, set $y = z = 0$ to get the successive iteration followed by

Iteration I: $x = \frac{4 - 3 \times 0 - 0}{5} = 0.8000$

$$y = \frac{-4+3 \times 0.8000 - 2 \times 0}{6} = -0.2667$$

$$z = \frac{3 + 0.8000 + 0.2667}{3} = 1.3556$$

Iteration 2: $x = \frac{4 - 3 \times (-0.2667) - 1.3556}{5} = 0.6889$

$$y = \frac{-4+3 \times 0.6889 - 2 \times 1.3556}{6} = -0.7741$$

$$z = \frac{3 + 0.6889+0.7741}{3} = 1.4877$$

Also, absolute relative error formula is given by

$$E = \left|\frac{x_{n+1} - x_n}{x_{n+1}}\right| \times 100\%$$

$\therefore$ The absolute relative error of the solution of the system in second order is

$$E(x) = \left|\frac{x_2 - x_1}{x_2}\right| \times 100\% = 16.13\%$$

$$E(y) = \left|\frac{y_2 - y_1}{y_2}\right| \times 100\% = 65.5\%$$

$$E(z) = \left|\frac{z_2 - z_1}{z_2}\right| \times 100\% = 8.879\%$$

Note: *In each table, calculator is fined on four decimal places* **[*Question No: (2) and (3)*]**

2. Solve the following system of equations by Gauss -Seidel method.

a) $5x_1 + x_2 = 7,\ x_1 + 6x_2 = 13$

Sol^n: Given system is

$$5x_1 + x_2 = 7$$
$$x_1 + 6x_2 = 13 \quad \ldots\ (1)$$

System (1) is diagonally dominant, we can apply Gauss-Seidel method. Therefore the system (i) is rewrite as $x_1 = \frac{7 - x_2}{5}$ and $x_2 = \frac{13 - x_1}{6}$ …… (3)

Initially, we set $x_2 = 0$ in (2), we get first iteration

$$x_1 = \frac{7 - 0}{5} = \frac{7}{5} = 1.4$$

and $x_1 = \frac{7}{5}$ in (3), we get

$$x_2 = \frac{13 - \frac{7}{5}}{6} = \frac{29}{15}$$

We continue this iteration by using table

Iteration	x_1	x_2
1	1.4	1.9333
2	1.0133	1.9978
3	1.0004	1.9999
4	1	2
5	1	2

$\therefore$ $x_1 = 1, x_2 = 2$

b) $3x_1 + x_2 = 13,\ x_1 - 6x_2 = 17$

Sol^n: Given system is

$$3x_1 + x_2 = 13$$
$$x_1 + 6x_2 = 17 \quad \ldots\ldots\ (1)$$

System (i) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as $x_1 = \frac{13 - x_2}{3}$ and $x_2 = \frac{17 - x_1}{-6}$

$$\boxed{\Rightarrow x_1 = \frac{13 - x_2}{3},\ x_2 = \frac{x_1 - 17}{6}}$$

Initially, we set $x_2 = 0$ to get the successive iteration in table below:

Iteration	x_1	x_2
1	4.3333	–2.1111
2	5.0370	–1.9938
3	4.9979	– 2.0003
4	5.0001	–2.0000
5	5.0000	–2.0000

$\therefore \quad x_1 = 5, x_2 = -2.$

c) $\mathbf{6x_1 - x_2 = 14,\ 2x_1 + 5x_2 = 26}$

Sol^n: Given system is

$$6x_1 + x_2 = 14$$
$$2x_1 + 5x_2 = 26 \quad \ldots\ldots (1)$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as $x_1 = \frac{14 - x_2}{6}$ and $x_2 = \frac{26 - 2x1}{5}$

Initially, we set $x_2 = 0$ to get the successive iteration in table below:

Iteration	x_1	x_2
1	2.3333	4.2667
2	3.0444	3.9822
3	2.9970	4.0012
4	3.0007	3.9999
5	3.0000	4.0000

$\therefore \quad x_1 = 3$ and $x_2 = 4$

d) $\mathbf{5x - 2y = -8,\ 3x - 10y = 4}$

Sol^n: Given system is

$$5x + 2y = -8$$
$$3x + 10y = 4 \quad \ldots. (1)$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$\boxed{\therefore x = \frac{2y - 8}{5},\ y = \frac{3x - 4}{10}}$$

Initially, we set $y = 0$ to get the successive iteration in table below:

Iteration	x	y
1	– 1.6000	– 0.8800
2	– 1.9520	– 0.9856
3	– 1.9942	– 0.9983
4	– 1.9993	– 0.998
5	– 1.9999	– 1.0000
6	– 2.0000	– 1.0000

$\therefore \quad x = -2, y = -1$

e) $\mathbf{3x + 7y = -5,\ 2x - y = 8}$

Give system is

$$3x + 7y = -5$$
$$2x - y = 8 \quad \ldots\ldots (1)$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore, the system (i) can be rewrite as $\boxed{x = \frac{8+y}{2}, y = \frac{-3x-5}{y}}$

Initially, we set y = 0 to get the successive iteration in table below:

Iteration	x	y
1	4.000	– 2.4286
2	2.7857	– 1.9082
3	3.0459	– 2.0197
4	2.9902	– 1.9958
5	3.0021	– 2.0009
6	2.9995	– 1.9998
7	3.0001	– 2.0000
8	3.0000	– 2.0000

$\therefore \quad x = 3, y = -2$

f) $5x_1 + 12x_2 = 26, 9x_1 + 4x_2 = -6$

Sol^n: Given system is

$$5x_1 + 12x_2 = 26$$
$$9x_1 + 4x_2 = -6$$

Rewrite the system so that it would be diagonally dominant as

$$9x_1 + 4x_2 = -6$$
$$5x_1 + 12x_2 = 26 \quad \ldots\ldots\ldots (1)$$

System (1) is diagonally dominant. We can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$\boxed{x_1 = \frac{-6-4x_2}{9}, x_2 = \frac{26-5x_1}{12}}$$

Initially, we set $x_2 = 0$ to get the successive iteration in table below:

Iteration	x	y
1	– 0.6667	2.444
2	– 1.7531	2.8971
3	– 1.9543	2.9809
4	– 1.9915	2.9965
5	– 1.9984	2.9993
6	– 1.9997	2.9999
7	– 1.9999	3.0000
8	– 2.0000	3.0000

$\therefore \quad x_1 = -2$ and $x_2 = 3$

3. Solve the following system of equations by Gauss seidel method.

a) $3x_1 + x_2 + x_3 = 5,\ x_1 - 4x_2 + x_3 = -2,\ x_1 + x_2 - 3x_3 = -1$

Sol^n: Given system is

$$\begin{aligned} 3x_1 + x_2 + x_3 &= 5 \\ x_1 - 4x_2 + x_3 &= -2 \quad \ldots\ldots\ldots (1) \\ x_1 + x_2 - 3x_3 &= -1 \end{aligned}$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$x_1 = \frac{5 - x_2 - x_3}{3},\ x_2 = \frac{2 + x_1 + x_3}{4} \text{ and } x_3 = \frac{1 + x_1 + x_2}{3}$$

Initially, set $x_2 = x_3 = 0$ to get the successive iteration in table below:

Iteration	x_1	x_2	x_3
1	1.6667	0.9167	1.1944
2	0.9630	1.0394	1.0008
3	0.9866	0.9968	0.9945
4	1.0029	0.9993	1.0007
5	1.0000	1.0002	1.0000
6	0.9999	1.0000	1.0000
7	1.0000	1.0000	1.0000

$\therefore \quad x_1 = 1, x_2 = 1, x_3 = 1$

b) $5x_1 + 3x_2 - x_3 = 15,\ 2x_1 - 4x_2 + x_3 = -4,\ x_1 - x_2 - 3x_3 = -13$

Sol^n: Given system is

$$\begin{aligned} 5x_1 + 3x_2 - x_3 &= 15 \\ 2x_1 - 4x_2 + x_3 &= -4 \quad \ldots\ldots\ldots (1) \\ x_1 - x_2 - 3x_3 &= -13 \end{aligned}$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$x_1 = \frac{15 - 3x_2 + x_3}{5},\ x_2 = \frac{4 + 2x_1 + x_3}{4},\ x_3 = \frac{13 + x_1 - x_2}{3} \quad \ldots\ldots\ldots (4)$$

Initially, set $x_2 = x_3 = 0$ to get the successive iteration in table below:

Iteration	x_1	x_2	x_3
1	3	2.5	4.5
2	2.4	3.3250	4.0250
3	1.8100	2.9113	3.9663
4	2.0465	3.0148	4.0106
5	1.9932	2.9993	3.9980
6	2.0000	2.9995	4.0002
7	2.0003	3.0002	4.0000
8	1.9999	3.0000	4.0000
9	2.0000	3.0000	4.0000

$\therefore \quad x_1 = 2, x_2 = 3$ and $x_3 = 4$

c) $4x_1 + x_2 + x_3 = 7,\ 2x_1 - 5x_2 + 2x_3 = 1,\ x_1 - x_2 + 3x_3 = 6$

Sol^n: Given system is

$$\begin{aligned} 4x_1 + x_2 + x_3 &= 7 \\ 2x_1 - 5x_2 + 2x_3 &= 1 \quad \ldots\ldots\ldots (1) \\ x_1 - x_2 + 3x_3 &= 6 \end{aligned}$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$x_1 = \frac{7 - x_2 - x_3}{4},\ x_2 = \frac{-1 + 2x_1 + 2x_3}{5} \text{ and } x_3 = \frac{6 - x_1 + x_2}{3} \quad \ldots\ldots\ (4)$$

Initially, set $x_2 = x_3 = 0$ to get the successive iteration in table below:

Iteration	$\mathbf{x_1}$	$\mathbf{x_2}$	$\mathbf{x_3}$
1	1.75	0.5	1.5833
2	1.2292	0.9250	1.8986
3	1.6441	0.9771	1.9777
4	1.0113	0.9956	1.9948
5	1.0024	0.9989	1.9988
6	1.0006	0.9998	1.9997
7	1.0001	0.9999	1.9999
8	1.0000	1.0000	2.0000

$\therefore \quad x_1 = 1, x_2 = 1, x_3 = 2$

d) $3x + y - z = 2,\ 2x - 5y + z = 20,\ x - 3y - 8z = 3$

Sol^n: Given system is

$$\begin{aligned} 3x + y - z &= 2 \\ 2x - 5y + z &= 20 \\ x - 3y - 8z &= 3 \end{aligned} \quad \ldots\ldots\ (1)$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system is rewrite as $x = \frac{2 - y + z}{3}$, $y = \frac{-20 + 2x + z}{5}$ and $z = \frac{-3 + x - 3y}{8}$

Initially, set $y = z = 0$ to get the successive iteration in table below:

Iteration	**x**	**y**	**z**
1	0.6667	– 3.7333	1.1083
2	2.2806	– 2.8661	0.9849
3	1.9503	– 3.0229	1.0024
4	2.0084	– 2.9962	0.9996
5	1.9986	– 3.0006	1.0001
6	2.0002	– 2.9999	1.0000
7	2.0000	– 3.0000	1.0000

$\therefore \quad x = z, y = -3, z = 1$

e) $y - 2z = 5,\ x - 4y = 2,\ 5x - y + z = 27$

Sol^n Since the system is not diagonally dominant, the system is rewrite so that it would be diagonally dominant as

$$\begin{aligned} 5x - y + z &= 27 \\ x - 4y + 0z &= 2 \\ 0x + y - 2z &= 5 \end{aligned} \quad \ldots\ldots\ (1)$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite as

$$x = \frac{2y + y - z}{5},\ y = \frac{-2 + x}{4} \text{ and } z = \frac{-5 + y}{2}$$

Initially, set y = z = 0 to get the successive iteration in table below:

Iteration	x	y	z
1	5.4	0.85	– 2.0750
2	5.9850	0.9963	– 2.0019
3	5.9996	0.9999	– 2.0000
4	6.0000	1.0000	– 2.0000

$\therefore$ x = 6, y = 1, z = – 2

f) 2x – 5y = 19, x + 5y – 8z = – 21, 4x – z = 7

Sol^n Since the system is not diagonally dominant, we rewrite the system so that it would be diagonally dominant

$$\begin{aligned} 4x - z &= 7 \\ 2x - 5y &= 19 \quad \ldots\ldots\ldots\ldots (1) \\ x + 5y - 8z &= -21 \end{aligned}$$

System (1) is diagonally dominant, can apply Gauss-Seidel method. Therefore the system (i) is rewrite in the form

$$x = \frac{z+7}{4}, y = \frac{2x-19}{5} \text{ and } z = \frac{x+5y+21}{8}$$

Initially, set z = 0 to get the successive iteration in table below:

Iteration	x	y	z
1	1.75	– 3.1	0.9063
2	1.9766	– 3.0094	0.9912
3	1.9978	– 3.0009	0.9992
4	1.9998	– 3.0001	0.9999
5	2.0000	– 3.0000	1.0000

$\therefore$ x = 2, y = – 3, z = 1

4. Solve the following equations

a) $2.003x_1 + x_2 = 50.06, 2x_1 + x_2 = 50$

also discuss the ill-conditioning when the coefficient of x_1 in first equation is changed to 2.0004

Sol^n: Given system is

$2.003x_1 + x_2 = 50.06$ … (i)

$2x_1 + x_2 = 50$ …(ii)

Solving (i) and (ii) we get $(x_1, x_2) = (20, 10)$

If we change the coefficient of x_1 of equation (i) to 2.0004, the system becomes

$2.0004\, x_1 + x_2 = 50.06$ … (iii)

$2x_1 + x_2 = 50$ …(iv)

Solving (iv) and (v) we get $(x_1, x_2) = (150, -250)$

Which shows that small change in the coefficient of x_1 yields large change in the solution vector

$\therefore$ The system is ill-conditioned.

b) $2x_1 + 3x_2 = 58, 2x_1 + 3.005x_2 = 58.08$

also discuss the ill-conditioning when the coefficient of x_2 in first equation is changed to 3.0005

Sol^n: Given system is

$2x_1 + 3x_2 = 58$ … (i)

$2x_1 + 3.005x_2 = 58.08$ …(ii)

Solving (i) and (ii) we get $(x_1, x_2) = (5, 16)$

If we change the coefficient of x_1 of equation (i) to 3.0005, the system becomes

$2x_1 + 3x_2 = 58$ ... (iii)

$2x_1 + 3.0005x_2 = 58.08$ ...(iv)

Solving (iv) and (v) we get $(x_1, x_2) = (-211, 160)$

Which shows that small change in the coefficient of x_1 yields large change in the solution vector

$\therefore$ The system is ill-conditioned.

c) Examine the ill-conditioned of the following equations

i) $211x - 125x = 47$
$405x - 240y = 90$

ii) $21x - 16x = 4$
$3x + 2y = 2$

Sol^n: i) Here, $211x - 125x = 47$

$405x - 240y = 90$

The determinant of input matrix is

$\begin{vmatrix} 211 & -125 \\ 405 & -240 \end{vmatrix} = -15$ which is not small in the sense of magnitude

So,Let us change given system into $y = mx + c$ form as

$1.688x - y = 0.357$ ..... (i)

$1.6875x - y = 0.376$ ..... (ii)

Again, $\begin{vmatrix} 1.688 & -1 \\ 1.6875 & -1 \end{vmatrix} = -0.0005$ which is very small (magnitude sense). Thus, system is ill-conditioned

Thing to remember

If two lines

$a_1x + b_1y = c_1$

$a_2x + b_2y = c_2$

are almost parallel then use determinant technique but it is not generalized idea. For getting small magnitude of system, we may change given system into $y = mx + c$ form and then determinant

ii) Here, $21x - 16y = 4$

$3x + 2y = 2$

Solving the system we get $x = 0.444$, $y = 0.333$

But, if we slightly change in the coefficient of x in first equation to 21.001 then system becomes

$21.001x - 16y = 4$

$3x + 2y = 2$

solving the above system we get $x = 0.44434$, $y = 0.33348$

This shows that small change in input coefficient does not yield large change in solution. Also, both solutions are almost same and closed. Thus the system is well-conditioned.

Hint and Solution of MCQ's

1. The system of equations

$a_1x + b_1y = c_1$ and

$a_2x + b_2y = c_2$ such that ($a_1 > b1$) will be convergent if the system is diagonally dominant

2. The system of equations

$a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ can be solved if $|a_1| > |b_1|$ and $|b_2| > |a_2|$

3. The system

$x - 2y + 5z = 8$ ..... (i)

$3x - y - z = 2$..... (ii)

$5x + 8y - 2z = 10$..... (iii)

will be diagonally dominant if order of equations will be ii, iii, i

i.e. $3x - y - z = 2$

$5x + 8y - 2z = 10$

$x - 2y + 5z = 8$ such that

diagonal		non-diagonals
$\|3\| = 3$	>	$\|-1\| + \|-1\| = 2$
$\|8\| = 8$	>	$\|5\| + \|-2\| = 7$
$\|5\| = 5$	>	$\|1\| + \|-2\| = 3$

4. To solve the system

$3x - y = 10$ ..... (i)

$8x + 12y = 15$ .....(ii), using Gauss-Seidel the initial values of (x, y) = (0, 0).

5. The system, $3x - 2y = 1$ ..... (i) and

$x + 3y = 3$ ..... (ii) are diagonally dominant, so the system is convergent.

6. Since, the system is diagonally dominant, which is rewrite as

$$x = \frac{12 - 2y}{3} \text{ ..... (i)}$$

$$y = \frac{4 + x}{2} \text{ ..... (ii)}$$

Initially, $y = 0$

Iteration I: $x = \frac{12 - 0}{3} = 4$

$$y = \frac{4 + 4}{2} = 4$$

Iteration II: $x = \frac{12 - 2.4}{3} = \frac{4}{3}$

$$y = \frac{4 + \frac{4}{3}}{2} = \frac{16}{6} = \frac{8}{3}$$

$\therefore \quad (x, y) = \left(\frac{4}{3}, \frac{8}{3}\right)$